Customer Lifetime Value

UCI Online Retail Dataset

A five-question analysis of customer behavior, retention, and revenue across 14 cohorts.

J. Glenn · Analytics Portfolio · 2026

Project Overview

Q1 — **Customer Profile**
Who are our customers and how are they distributed?

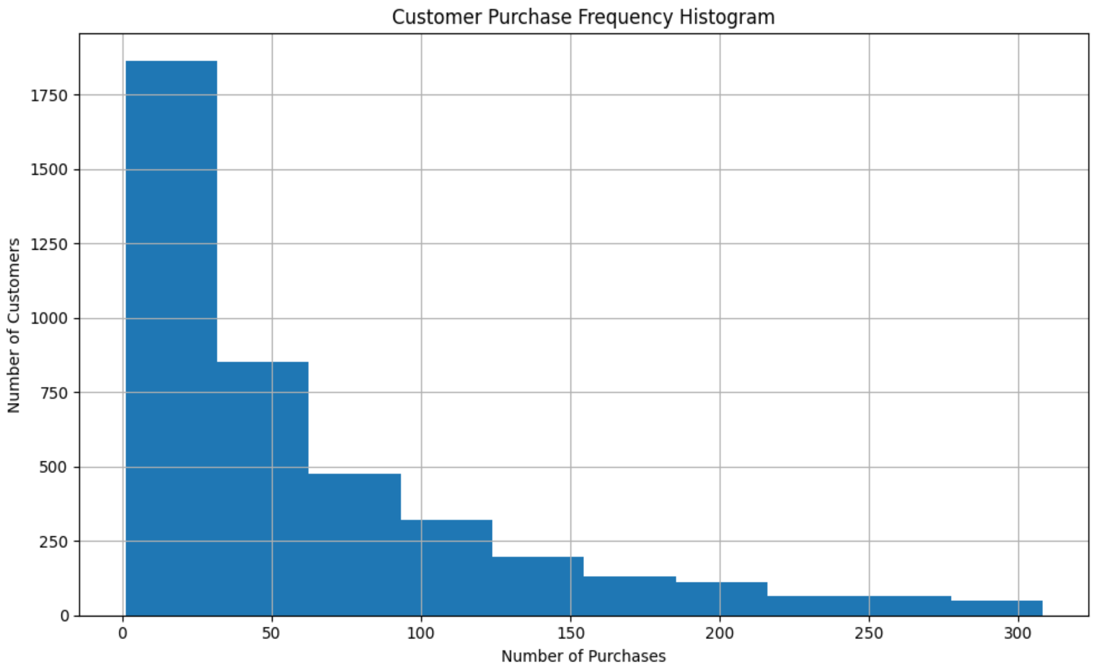
Q2 — **Buying Habits**
How frequently do they buy and what do they spend?

Q3 — **Geographic Reach**
Which markets generate the most revenue?

Q4 — **Retention**
Do customers come back after their first purchase?

Q5 — **Lifetime Value**
What does each cohort spend over their lifetime?

Who Are Our Customers?



4,338
Total Customers

41
Median Purchases

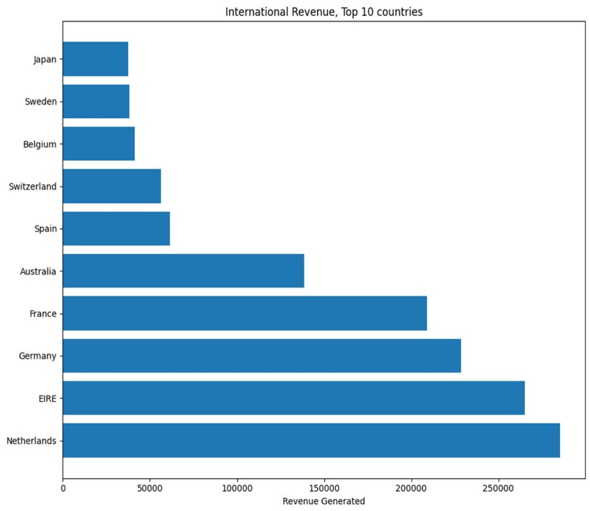
£17.76
Median Avg Spend

£668
Median Total Spend

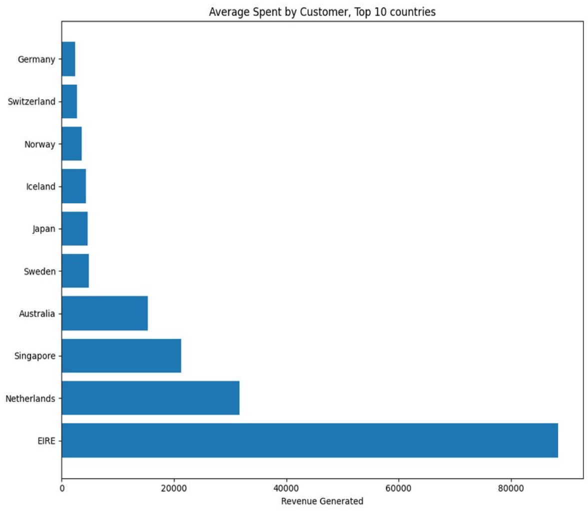
Most customers make fewer than 50 purchases. A small high-frequency segment skews the distribution, consistent with wholesale buyer behavior identified in cleaning.

Q3 · Geographic Segmentation

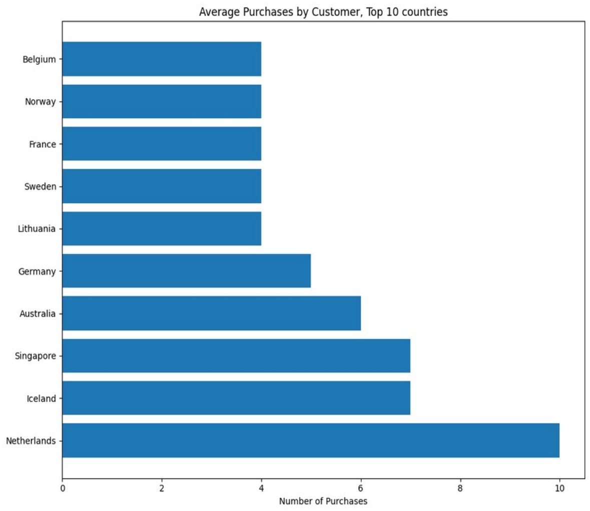
Where Does Revenue Come From?



Total Revenue



Avg Spend / Customer

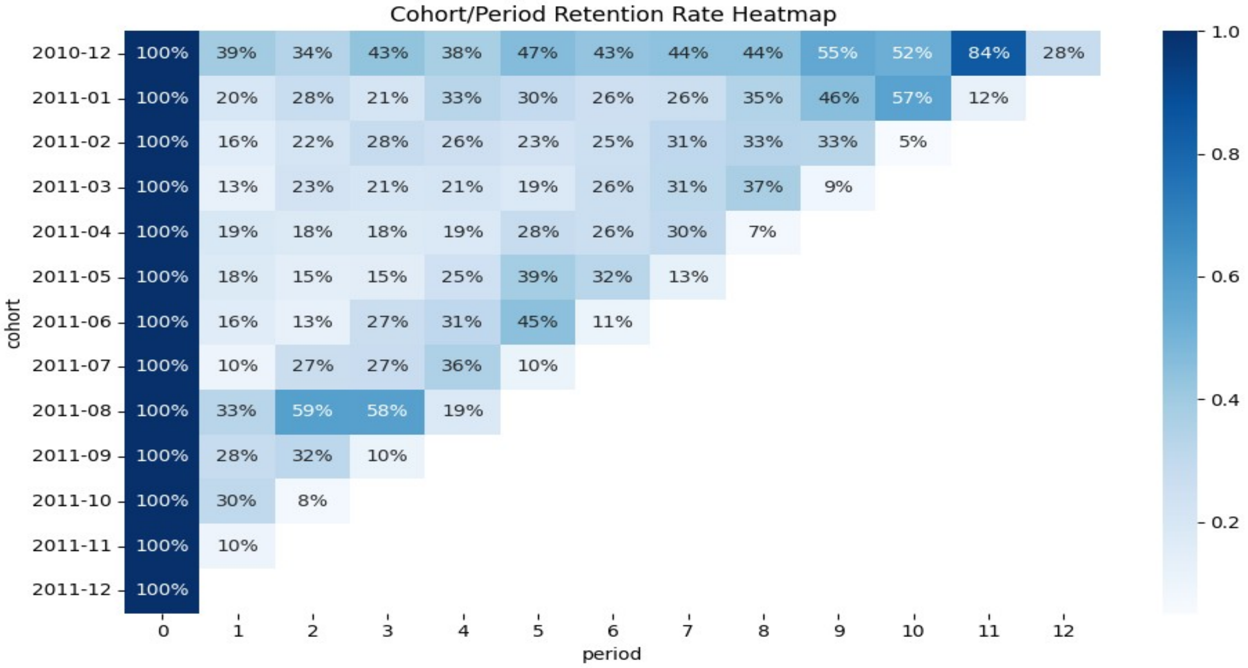


Avg Purchases / Customer

Netherlands leads on total revenue and purchase frequency. EIRE dominates average spend per customer, driven by 3 high-volume wholesale accounts, not broad market depth.

Q4 · Cohort Retention

Do Customers Come Back?



2010-12

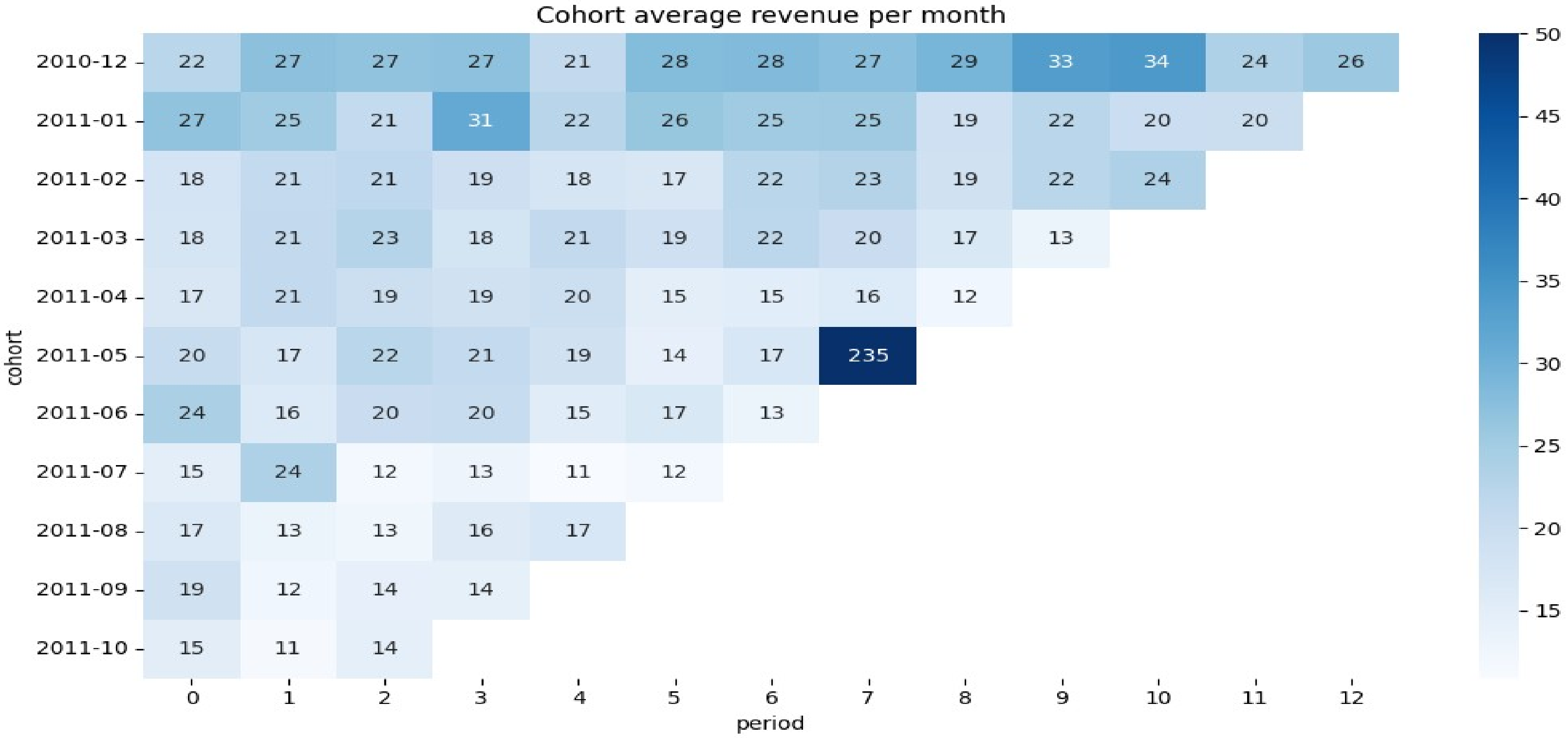
Strongest cohort with highest retention rates across all periods

Later Cohorts

Compressed, similar patterns with cause indeterminate from this dataset

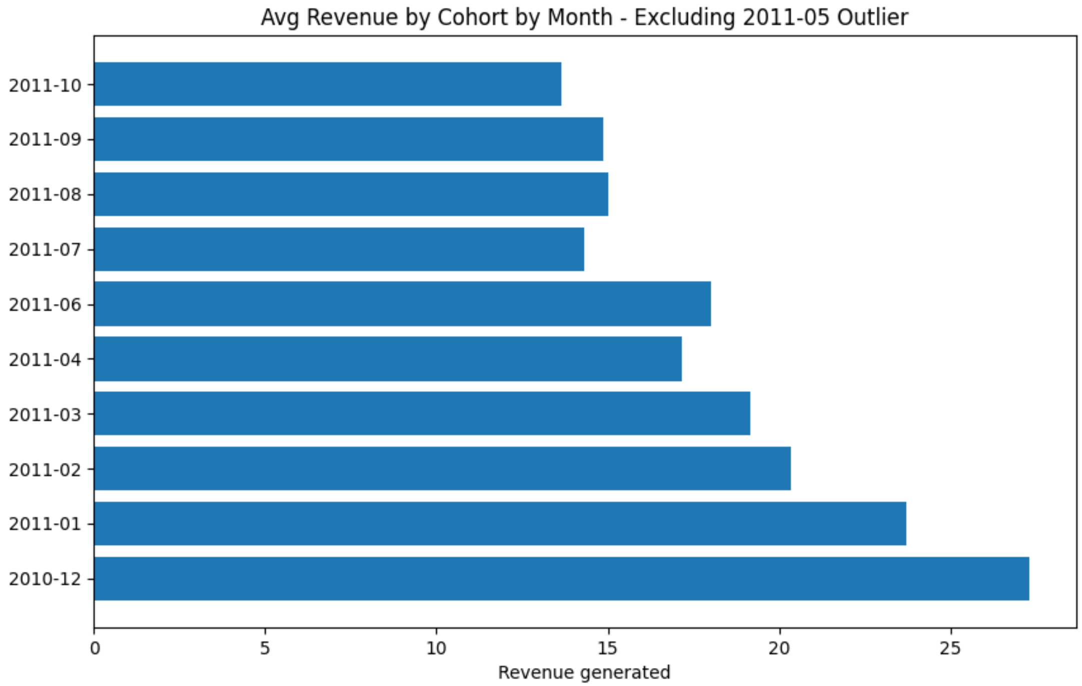
Retention rates are month-over-month within each cohort. Each cell is directly comparable across cohorts at the same period.

Average Revenue Per Customer by Cohort & Month



Cells show average £ revenue per customer. Color scale capped at £50 for readability. 2011-05 period 7 outlier (£235) reflects a single high-value transaction event.

Earlier Cohorts Spend More



£27

2010-12 avg spend
(highest cohort)

£14

2011-07 avg spend
(lowest cohort)

Cohorts with 3+ months of data only. 2011-05 excluded (single transaction outlier distorts mean).

Key Findings

What the Data Reveals

01 Early Cohorts Are Your Best Customers

The 2010-12 cohort leads on both retention and average spend (£27/month). Customer quality, measured by transaction value, has declined across the observation period.

02 Geographic Concentration Carries Risk

Netherlands and EIRE dominate revenue, but EIRE's figures are driven by 3 wholesale accounts. Volume and customer quality are different stories.

03 Retention Drops Sharply After Month 1

Across all cohorts, the largest retention drop occurs between period 0 and period 1. Winning a customer's second purchase is the critical inflection point.

04 Later Cohort Patterns Are Indeterminate

Cohorts from 2011-07 onward show compressed, similar behavior. Whether this is market saturation, seasonality, or data truncation cannot be determined from this dataset alone.

Implications · What the Data Suggests

Where to Look Next

I

Invest in Australia

Finding 02 — Geographic Concentration Carries Risk

Australia ranks 4th or 5th across all three geographic metrics: total revenue, average spend per customer, and purchase frequency. That consistency signals a real, engaged customer base. A modest increase in market investment could accelerate growth in a market that is already responding.

II

Investigate the 2011-08 Cohort

Finding 04 — Later Cohort Patterns Are Indeterminate

The August 2011 cohort shows 59% and 58% retention at periods 2 and 3 — well above surrounding cohorts. Something drove that behavior. Identifying whether it was a promotion, a product launch, or a channel change could provide a repeatable playbook for improving retention across all cohorts.

III

Replace Email with Direct Outreach After First Purchase

Finding 03 — Retention Drops Sharply After Month 1

The largest retention drop occurs between period 0 and period 1. A phone-based contact campaign after first purchase, not email, builds a personal relationship before the customer goes dormant. The goal is to make the second purchase feel like a natural next step rather than a marketing event.